

SHETH L.U.J. AND SIR M.V. COLLEGE

AIM: Implement CSMA/CA protocol simulation and evaluate collision avoidance mechanisms.

Setting Up the Wireless Topology

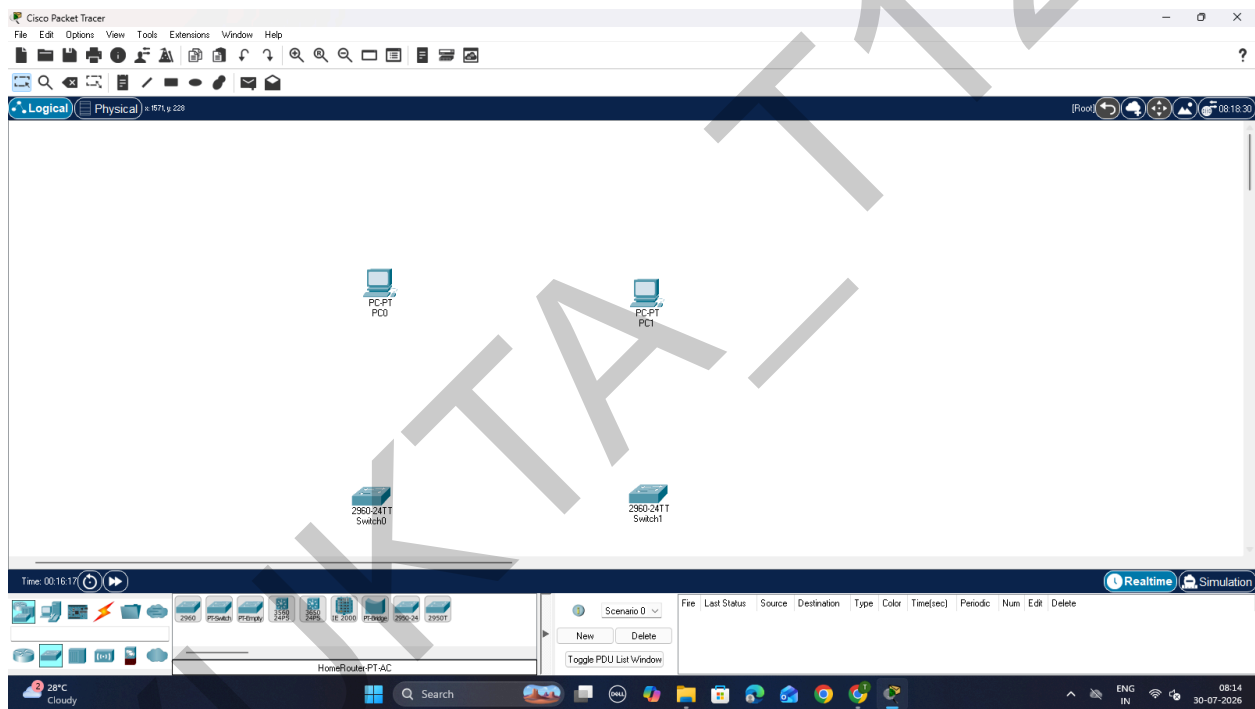
Add a Wireless Router (e.g., Model WRT300N or AccessPoint-PT) to the workspace.

Place two or more Wireless Laptops or PCs near the router.

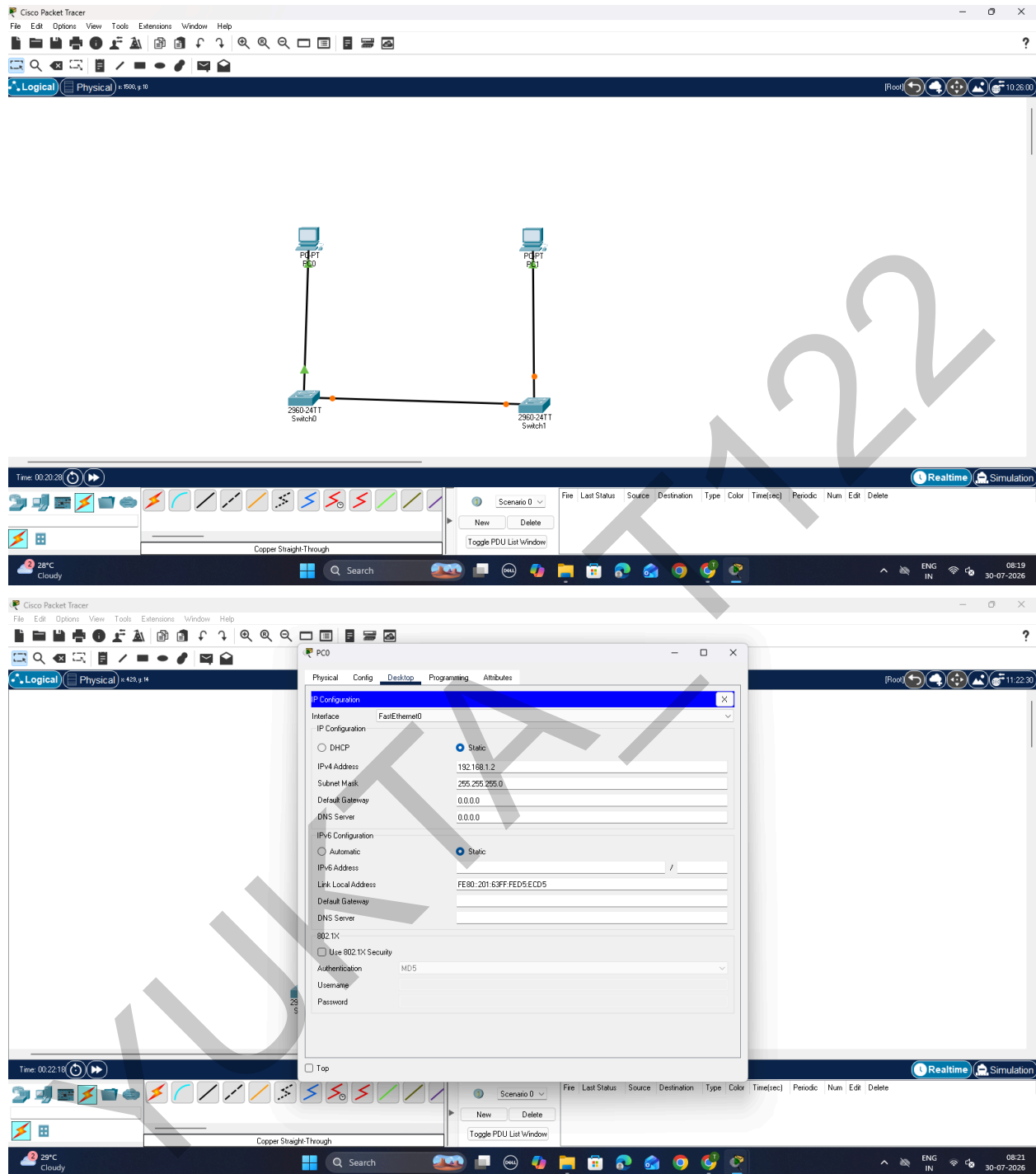
Click a laptop, go to the Physical tab, turn off the power, replace the standard fast-ethernet module with a wireless module (e.g., WMP300N or Linksys-WPC300N), and turn the power back on.

Go to the Config or Desktop -> IP Configuration tab on each laptop and assign static IPv4 addresses (e.g., 192.168.0.2 and 192.168.0.3 with subnet 255.255.255.0).

Verify association by checking that a visual wireless link line appears between the laptops and the wireless router.

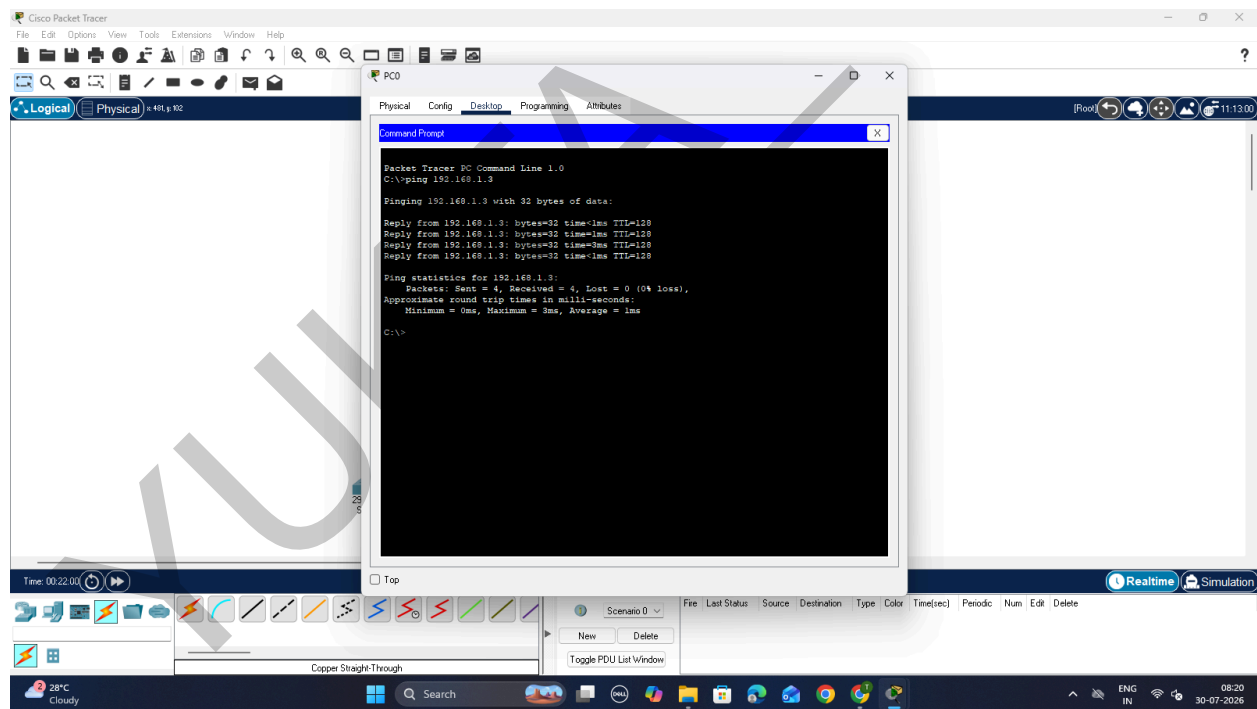
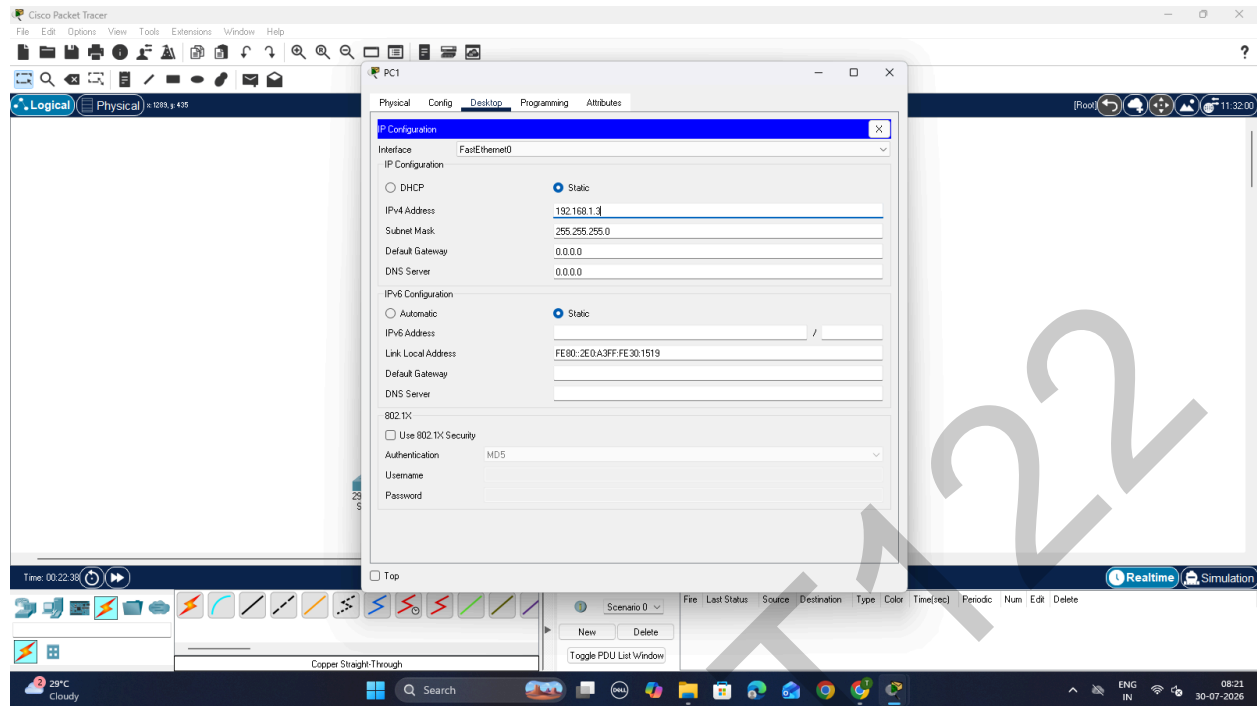


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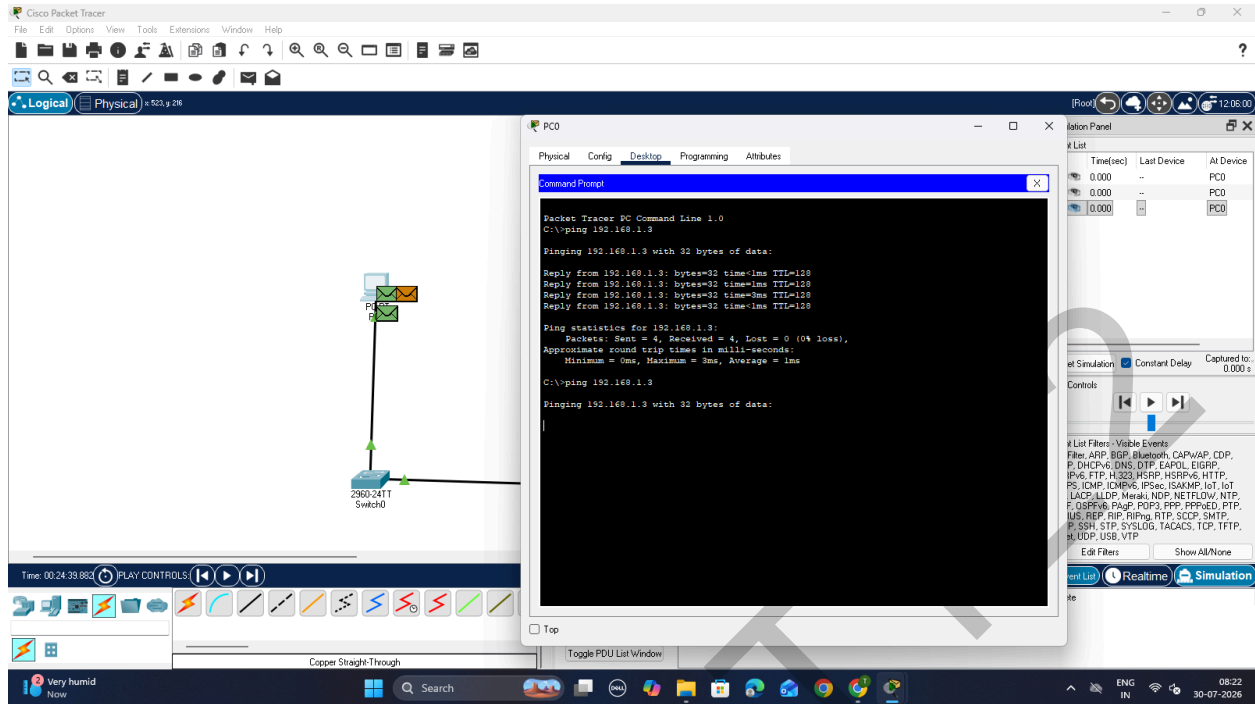
YUKTA SONAWANE T122
WSN PRACTICAL NO : 4

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Configuring and Running the Simulation

Switch from Real-Time mode to Simulation mode using the bottom panel or the stopwatch icon.

Open Edit Filters in the simulation event list and select ICMP and 802.11 or ARP/MAC-related protocols if visible.

Use the Add Simple PDU (envelope tool) to send a packet from Laptop 0 to Laptop 1.

Click Capture / Forward to step through the frame transmission process packet by packet.

Observe how the wireless medium coordinates access: nodes listen to the channel (Carrier Sense) and wait through random backoff/interframe spaces to prevent simultaneous **broadcasts (Collision Avoidance) before the Access** Point relays the frame.

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The top screenshot shows a network topology in Cisco Packet Tracer. It features two switches, 2960-24TT Switch0 and 2960-24TT Switch1, connected by a line. Each switch is connected to a PC, 830. The interface includes a menu bar, a toolbar, and a 'Simulation Panel' on the right. The 'Event List' table in the Simulation Panel is as follows:

Vis.	Time(sec)	Last Device	At Device
	0.000	--	PC0
	0.000	--	PC0
	0.000	--	PC0
	0.001	PC0	Switch0
	0.001	--	PC0

The bottom screenshot shows a similar network topology. The switches are connected to the PCs, and a 'Copper Straight-Through' patch cable is visible. The 'Simulation Panel' on the right shows a different 'Event List' table:

Vis.	Time(sec)	Last Device	At Device
	0.000	--	PC0
	0.000	--	PC0
	0.000	--	PC0
	0.001	PC0	Switch0
	0.001	--	PC0
	0.002	PC0	Switch0
	0.002	Switch0	Switch1
	0.003	Switch0	Switch1
	0.003	Switch1	PC1
	0.004	Switch1	PC1